LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
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TASK 2 — HTML5 CANVAS AND CSS STYLING

1. TITLE

Implement HTML5 Canvas Graphics and Design a Styled Web Page Using CSS

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2. AIM

To understand and implement HTML5 Canvas for drawing basic graphical elements and CSS for styling and designing a structured web page.

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3. OBJECTIVE

The objectives of this task are:

- To understand the HTML5 <canvas> element.
- To obtain the 2D drawing context using JavaScript.
- To draw basic shapes using Canvas.
- To draw a rectangle, circle, and line.
- To display text using Canvas.
- To understand external CSS styling.
- To apply fonts, colors, spacing, borders, and shadows.
- To create a navigation bar using CSS.
- To design card-based content using Flexbox.
- To apply hover effects to links and buttons.
- To create a visually styled web page.

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4. THEORY

HTML5 Canvas

The HTML5 Canvas element provides a drawing area that can be controlled using JavaScript. It can be used to create graphics, shapes, text, animations, and other visual elements.

The <canvas> element defines the drawing area, while JavaScript is used to perform the actual drawing.

The getContext("2d") method obtains the 2D drawing context, which provides methods for drawing shapes and text.

In this task:

- fillRect() is used to draw a rectangle.
- arc() is used to create a circular path.
- stroke() is used to draw the line.
- fillText() is used to display text.
- fillStyle is used to set the fill color.
- font is used to define the text font.

CSS

CSS (Cascading Style Sheets) is used to control the appearance and layout of HTML elements.

External CSS can be connected to an HTML page using the <link> element.

In this task, CSS is used for:

- Font styling
- Background gradient
- Text color
- Navigation styling
- Flexbox layout
- Cards
- Borders
- Rounded corners
- Box shadows
- Button styling
- Hover effects
- Footer styling

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5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare the HTML5 document type using <!DOCTYPE html>.
2. Create the <head> section with character encoding, viewport settings and the page title.
3. Write internal CSS3 styling inside a <style> block for the body, header, navigation, cards, canvas section and footer.
4. Apply a linear-gradient() background to the <body> using CSS3.
5. Create a <header> section displaying the webpage title and a short description, styled with borders, border-radius and box-shadow.
6. Create a <nav> section with Home, HTML5, CSS3 and Canvas links, styled as rounded buttons with a hover effect.
7. Create a container <section> with three cards (HTML5, CSS3, Canvas) using <div class="card">, each with a heading, paragraph and button.
8. Apply CSS3 transitions and a hover effect (translateY) to the cards, and a hover/scale effect to the buttons.
9. Create a canvas section containing a heading and a <canvas> element with a defined width and height.
10. Write a <script> block that gets the 2D drawing context of the canvas using getContext("2d").
11. Use fillRect() to draw a filled rectangle on the canvas.
12. Use beginPath(), arc() and fill() to draw a filled circle on the canvas.
13. Use beginPath(), moveTo(), lineTo() and stroke() to draw a straight line on the canvas.

14. Use font, fillStyle and fillText() to draw text (the student's name) on the canvas.
15. Create a <footer> section displaying the copyright text.
16. Save the HTML file and open it in a web browser.
17. Verify that the styling, hover effects and canvas drawing are displayed correctly.
18. Record the output using screenshots.

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6. PROGRAM

MyStyledWebpage.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Styled Webpage</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        /* Gradient Background */
        body {
            font-family: 'Trebuchet MS', sans-serif;
            background: linear-gradient(135deg, #89CFF0, #E0BBE4, #B5EAD7);
        }
        /* Header Section */
        header {
            background: #1F618D;
            color: white;
            text-align: center;
            padding: 35px;
            margin: 15px;
            border: 4px solid #154360;
            border-radius: 20px;
            box-shadow: 5px 5px 15px gray;
        }
        header h1 {
            font-family: Georgia;
            font-size: 50px;
            text-shadow: 3px 3px 5px black;
        }
        header p {
            font-family: Verdana;
            font-size: 22px;
            margin-top: 10px;
        }
        /* Navigation Section*/
        nav {
            background: #48C9B0;
            text-align: center;
            padding: 20px;
            margin: 15px;
            border: 3px solid #117A65;
            border-radius: 15px;
            box-shadow: 5px 5px 12px gray;
        }
        nav a {
            text-decoration: none;
            color: #2C3E50;
            background: white;
            padding: 12px 22px;
```

```
    margin: 15px;
    border: 2px solid #48C9B0;
    border-radius: 30px;
    font-size: 18px;
    font-weight: bold;
    transition: 0.4s;
}
nav a:hover {
    background: #8E44AD;
    color: white;
}
/* Cards */
.container {
    display: flex;
    justify-content: space-around;
    gap: 25px;
    margin: 30px;
}
.card {
    width: 30%;
    background: #FFF4CC;
    border: 4px solid #E67E22;
    border-radius: 20px;
    padding: 30px;
    text-align: center;
    box-shadow: 8px 8px 18px rgba(0, 0, 0, 0.3);
    transition: 0.4s;
}
.card:hover {
    background: #FEF9E7;
    transform: translateY(-10px);
}
.card h2 {
    color: #D35400;
    font-family: Georgia;
    font-size: 34px;
    text-shadow: 2px 2px 4px silver;
    margin-bottom: 20px;
}
.card p {
    font-family: Verdana;
    font-size: 18px;
    color: #34495E;
    line-height: 1.6;
    margin-bottom: 20px;
}
button {
    background: #7D3C98;
    color: white;
    border: 2px solid black;
    border-radius: 30px;
    padding: 12px 25px;
    font-size: 18px;
    cursor: pointer;
    transition: 0.4s;
}
button:hover {
    background: #F39C12;
    color: black;
    transform: scale(1.08);
}
/* Canvas Section */
.canvas-section {
    text-align: center;
    margin: 40px;
    padding: 20px;
    background: #EBF5FB;
    border: 3px solid #2874A6;
    border-radius: 20px;
```

```
            box-shadow: 6px 6px 15px gray;
        }
        .canvas-section h1 {
            font-size: 42px;
            color: #1F618D;
            text-shadow: 2px 2px 4px gray;
            margin-bottom: 20px;
        }
        canvas {
            background: white;
            border: 4px solid black;
            border-radius: 15px;
        }
        /* Footer Section*/
        footer {
            background: #273746;
            color: white;
            text-align: center;
            padding: 20px;
            margin: 20px;
            border: 3px solid black;
            border-radius: 15px;
            box-shadow: 5px 5px 12px gray;
            text-shadow: 2px 2px 4px black;
        }
    </style>
</head>
<body>
    <!--  PART B : CSS3 STYLING -->
    <header>
        <h1>My Styled Webpage</h1>
        <p>Learning HTML5 Canvas and CSS3 Styling</p>
    </header>
    <nav>
        <a href="#">Home</a>
        <a href="#">HTML5</a>
        <a href="#">CSS3</a>
        <a href="#">Canvas</a>
    </nav>
    <section class="container">
        <div class="card">
            <h2>HTML5</h2>
            <p>
                HTML5 provides semantic elements, forms, multimedia support and canvas for
creating modern webpages.
            </p>
            <button>Learn HTML</button>
        </div>
        <div class="card">
            <h2>CSS3</h2>
            <p>
                CSS3 enhances webpages using gradients, fonts, borders, colors and animations.
            </p>
            <button>Explore CSS</button>
        </div>
        <div class="card">
            <h2>Canvas</h2>
            <p>
                HTML5 Canvas allows drawing graphics like rectangles, circles, lines and text.
            </p>
            <button>View Demo</button>
        </div>
    </section>
    <!--  PART A : HTML5 CANVAS-->
    <section class="canvas-section">
        <h1>HTML5 Canvas</h1>
        <canvas id="myCanvas" width="600" height="300"></canvas>
    </section>
    <footer>
```

```
        <p>&copy; 2026 My Styled Webpage | HTML5 & CSS3 Practical</p>
    </footer>
    <script>
        var canvas = document.getElementById("myCanvas");
        var ctx = canvas.getContext("2d");
        /* Rectangle */
        ctx.fillStyle = "#27AE60";
        ctx.fillRect(40, 30, 150, 90);
        /* Circle */
        ctx.beginPath();
        ctx.arc(300, 75, 45, 0, 2 * Math.PI);
        ctx.fillStyle = "#F39C12";
        ctx.fill();
        /* Line */
        ctx.beginPath();
        ctx.moveTo(40, 190);
        ctx.lineTo(520, 190);
        ctx.strokeStyle = "#C0392B";
        ctx.lineWidth = 5;
        ctx.stroke();
        /* Name */
        ctx.font = "30px Georgia";
        ctx.fillStyle = "#1F618D";
        ctx.fillText("Suhas", 220, 260);
    </script>
</body>
</html>
```

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7. CODE EXPLANATION

HTML Elements:

Code / Tag	Explanation
<!DOCTYPE html>	Declares that the document uses HTML5.
<head>	Contains metadata, viewport settings, the page title and the internal <style> block.
<style>	Contains internal CSS3 rules used to style the entire webpage.
<header>	Defines the header section showing the webpage title and description.
<nav>	Contains the Home, HTML5, CSS3 and Canvas navigation links.
<section class="container">	Groups the three information cards (HTML5, CSS3, Canvas).
<div class="card">	Represents an individual card containing a heading, description and button.
<button>	Creates the Learn HTML, Explore CSS and View Demo action buttons.
<section class="canvas-section">	Groups the heading and the canvas drawing area.
<canvas>	Defines a drawing area on which graphics are rendered using JavaScript.
<script>	Contains JavaScript code used to draw shapes and text on the canvas.
<footer>	Defines the footer section containing the copyright information.

CSS3 Properties:

Code / Property	Explanation
* { margin, padding, box-sizing }	Resets default spacing and sets consistent box sizing for all elements.

Code / Property	Explanation
background (linear-gradient)	Applies a diagonal multi-colour gradient background to the page.
border-radius	Creates rounded corners on the header, nav, cards, canvas section, buttons and footer.
box-shadow	Adds a soft shadow effect around the header, nav, cards, canvas section and footer.
text-shadow	Adds a shadow effect behind text in the header and canvas section heading.
display: flex	Arranges the three cards in a row with equal spacing using .container.
transition	Produces a smooth animated change when hover styles are applied.
:hover	Changes background colour and applies a transform when the mouse is placed over nav links, cards and buttons.
transform: translateY() / scale()	Lifts a card upward or enlarges a button slightly on hover.
cursor: pointer	Shows a pointer cursor on interactive elements such as buttons.

JavaScript / Canvas API:

Code / Method	Explanation
getElementById("myCanvas")	Selects the <canvas> element from the document.
getContext("2d")	Obtains the 2D drawing context used to draw shapes and text.
fillStyle / fillRect()	Sets the fill colour and draws a filled rectangle.
beginPath() / arc() / fill()	Starts a new drawing path and draws a filled circle.
moveTo() / lineTo() / stroke()	Draws a straight line between two points on the canvas.
font / fillText()	Sets the text font and draws the student's name on the canvas.

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8. TEST CASES AND EXECUTION DATA

The following test cases were executed on the styled webpage and the results were recorded.

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a web browser	Webpage should be displayed with gradient background and styling	Webpage displayed correctly	Pass
TC-02	Hover over the Home, HTML5, CSS3 and Canvas nav links	Link background and text colour should change smoothly	Hover effect displayed	Pass
TC-03	Hover over the HTML5, CSS3 and Canvas cards	Card should lift upward with a smooth transition	Hover effect displayed	Pass
TC-04	Hover over the Learn HTML, Explore CSS and View Demo buttons	Button should change colour and scale slightly	Hover effect displayed	Pass

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-05	Check the HTML5 Canvas section	A rectangle, a circle, a line and the text "Suhas" should be drawn on the canvas	All shapes and text drawn correctly	Pass
TC-06	Resize or reload the browser window	Layout, gradients and canvas drawing should remain consistent	Layout displayed correctly	Pass

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9. OUTPUT

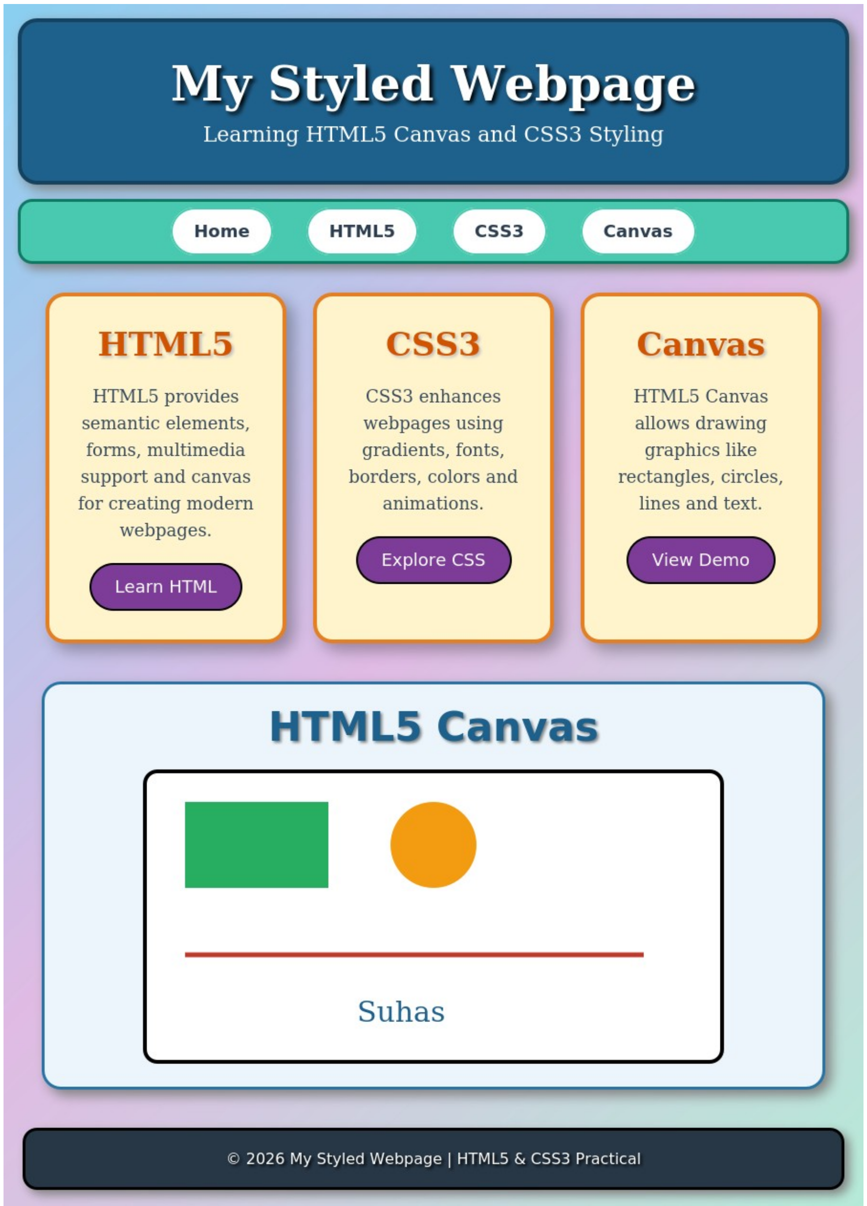

Figure 1: Output of the styled webpage showing header, navigation, HTML5/CSS3/Canvas cards, hover effects, and the HTML5 Canvas drawing (rectangle, circle, line and text).

10. RESULT

Thus, a visually attractive webpage was successfully designed using HTML5 and CSS3. CSS3 properties such as gradients, borders, border-radius, box-shadow, transitions and hover effects were used to style the header, navigation, cards, canvas section and footer. The HTML5 Canvas API was used with JavaScript to draw a rectangle, a circle, a line and text, and all test cases were executed successfully.